

SQL Security Log Investigation and Filtering

1. Project description

In this project, I used SQL queries to investigate potential security incidents and retrieve relevant information from organizational datasets. By applying filtering techniques such as AND, OR, NOT, and LIKE, I analyzed login activity and employee records to identify suspicious events, support investigations, and assist with security-related decision-making.

2. Key tasks

A) Retrieve after-hours failed login attempts

```
SELECT *  
FROM log_in_attempts  
WHERE login_time > '18:00'  
AND success = FALSE;
```

Purpose : Retrieved all failed login attempts that occurred outside normal business hours to support investigation of potential unauthorized access attempts.

B) Retrieve login attempts on specific dates

```
SELECT *  
FROM log_in_attempts  
WHERE login_time > '18:00'  
AND success = FALSE;
```

Purpose : Retrieved all failed login attempts that occurred outside normal business hours to support investigation of potential unauthorized access attempts.

C) retrieve login attempts on specific dates

```
SELECT *  
FROM log_in_attempts  
WHERE login_date = '2022-05-09'  
OR login_date = '2022-05-08';
```

Purpose : Investigated login activity surrounding a suspicious event by filtering records from specific dates.

D) Retrieve login attempts outside of Mexico

```
SELECT *  
FROM log_in_attempts  
WHERE NOT country LIKE 'MEX%';
```

Purpose : Identified login attempts originating outside Mexico to support geographic analysis of security events.

E) Retrieve employees in Marketing

```
SELECT *  
FROM employees  
WHERE department = 'Marketing'  
AND office LIKE 'EAST%';
```

Purpose : Retrieved employee systems requiring updates in the Marketing department located in the East building.

F) Retrieve employees in Finance or Sales

```
SELECT *  
FROM employees  
WHERE department = 'Finance'  
OR department = 'Sales';
```

Purpose : Identified systems belonging to Finance and Sales employees for department-specific security updates.

G) Retrieve all employees not in IT

```
SELECT *  
FROM employees  
WHERE NOT department = 'Information Technology';
```

Purpose : Filtered employee records to exclude IT personnel when planning security updates for other departments.

3. Skill demonstrated

- SQL querying,
- Data filtering,
- Security log analysis,
- Incident investigation,
- Security monitoring,
- Database querying,
- Pattern matching with LIKE
- Logical operators (AND, OR, NOT)

4, Summary

This project demonstrates the use of SQL to support cybersecurity investigations and operational security tasks. I analyzed login activity and employees records, using filtering techniques to identify suspicious events, investigate potential incidents, and retrieve information needed for security-related decisions. The project reinforced practical skills in security monitoring, log analysis, and data-driven investigations.